EviRefine: Evidence-Guided Coarse-to-Fine Multivariate Time-Series Imputation

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Working draft v0.1

Abstract

Multivariate time-series imputation requires recovering local trajectories from observations whose availability varies across time and variables. The central challenge is to integrate complementary contextual and patch-level information while retaining numerical consistency with visible measurements. We introduce EviRefine, an evidence-guided coarse-to-fine framework for multivariate time-series imputation. The framework comprises four components: temporal–variable context encoding, multi-source patch evidence retrieval, pointwise evidence-guided reconstruction, and observation-conditioned residual refinement. Context encoding captures temporal dynamics and cross-variable dependencies. Patch retrieval organizes current-window and training-history fragments into a common evidence bank, which is read separately at each target point to augment a contextual base estimate. Residual refinement subsequently combines this estimate with visible observations and observation-derived references to correct reconstruction errors. Experiments cover six datasets from diverse application domains. In paired three-seed studies against a capacity-matched context-only model, the evidence pathway reduces final MSE by 5.19% on ETTh1 and 2.39% on PAMAP2. Component, missing-condition and efficiency analyses further characterize how patch evidence and observation-conditioned refinement contribute to recovery.

1. Introduction

Multivariate time series record interacting processes in environmental monitoring, physiological sensing, industrial systems and financial markets. Missing observations interrupt these processes at different locations and durations, obscuring both local dynamics and relationships across variables. Recovering the missing values calls for a representation that integrates available information across these complementary dimensions.

Recent methods model incomplete observations through recurrent estimation, self-attention, diffusion and structured spatiotemporal representations [1, 2, 4, 7]. Temporal patches provide a useful unit for representing local trajectories [5], while variable-wise attention captures dependencies among measured processes [3, 6]. These advances motivate two complementary requirements for imputation: identifying informative fragments beyond a damaged local context, and converting contextual representations into numerical estimates consistent with visible measurements.

The availability of useful information changes with each missing pattern. A same-variable fragment can reveal a temporal continuation, a related variable can expose a shared transition, and a training-history fragment can supply a recurring trajectory. Their relevance also varies within a target patch: points close to a visible boundary and points deep inside a gap receive different local support. This motivates organizing patch-level evidence by source and reading it at the granularity of individual missing points.

Numerical correction provides a second opportunity for recovery. Once an initial estimate is available, the remaining visible observations anchor its level and local transitions. A dedicated refinement stage can use this joint representation to correct residual errors across the window. This leads to a coarse-to-fine design in which contextual reconstruction establishes an initial trajectory and observation-conditioned refinement improves its pointwise values.

Building on these considerations, we introduce EviRefine, an evidence-guided coarse-to-fine imputation framework with four components: (1) Temporal–Variable Context Encoding (TVCE), (2) Multi-Source Patch Evidence Retrieval (MPER), (3) Pointwise Evidence-Guided Reconstruction (PEGR), and (4) Observation-Conditioned Residual Refinement (OCRR). TVCE combines residual temporal convolution with variable-signature attention. MPER organizes temporal, cross-variable and training-history patches into a common bank. PEGR reads this bank separately for each target point and combines the readout with its query to correct a base estimate. OCRR then integrates the fixed coarse estimate with visible observations and reference features to produce the refined window.

Our contributions are summarized as follows:

- We introduce EviRefine, a coarse-to-fine framework that

integrates contextual reconstruction, explicit patch evidence and observation-conditioned correction for multivariate time-series imputation.
- We develop multi-source patch evidence retrieval and pointwise reconstruction, enabling target-specific access to complementary temporal, cross-variable and historical fragments through a common reading interface.
- We design an observation-conditioned residual refiner that combines temporal correction with visible-observation references. Experiments across six datasets and paired component studies evaluate the resulting framework and its recovery mechanisms.

2. Related Work

Multivariate time-series imputation. Deep imputation models exploit temporal dependencies and cross-variable relationships through different reconstruction mechanisms. BRITS uses bidirectional recurrent estimation [1], and SAITS employs diagonally masked self-attention [2]. CSDI models conditional distributions with score-based diffusion [7], while ImputeFormer introduces low-rank spatiotemporal attention [4]. Our framework integrates contextual estimation with explicit patch evidence and a subsequent observation-conditioned correction stage.

Temporal and cross-variable representation. PatchTST represents local trajectories using temporal patches [5]; iTransformer models dependencies between variate tokens [3]. T1 binds temporal feature channels to variable-attention heads [6]. TVCE adopts this channel–head geometry as its variable-signature attention building block. MPER and PEGR extend contextual reconstruction with source-organized fragments and pointwise reading, while OCRR revisits the resulting numerical estimate using visible observations.

Structured temporal modeling. TimeMixer++ combines multiple representation and mixing operations to capture temporal patterns and analyzes its components through ablation, representation, efficiency and sensitivity studies [9]. For imputation, we organize information around point context, patch evidence and observation-conditioned correction. Capacity-matched controls, shared refinement and mask-defined error strata connect these design choices to measured reconstruction behavior.

3. EviRefine

3.1. Problem Formulation and Architecture Overview

Let $\mathbf{X} \in \mathbb{R}^{L\times C}$ denote a multivariate window, $\mathbf{V}$ its native observation mask, and $\mathbf{M} \le \mathbf{V}$ the observations retained after artificial masking. Native missing values have no supervised target. We write $\mathbf{X}_{\text{obs}}$ for the visible values with all unavailable entries set to zero. The evaluation support and observed-value preservation are

$$\begin{aligned}\mathbf{E} &= \mathbf{V} \odot (\mathbf{1} - \mathbf{M}),\\ \widehat{\mathbf{X}} &= \mathbf{M} \odot \mathbf{X}_{\text{obs}} + (\mathbf{1} - \mathbf{M}) \odot f(\mathbf{X}_{\text{obs}}, \mathbf{M}).\end{aligned} \tag{1}$$

Batch indices are omitted. Inference uses the visible observations in the input window and the eligible training memory.

EviRefine has two imputation stages (Fig. 1): evidence-guided coarse recovery and observation-conditioned residual refinement. Coarse recovery produces an explicit estimate $\mathbf{X}^{\text{c}}$ in training-standardized coordinates. The refiner receives this estimate as a fixed input and returns $\widehat{\mathbf{X}}$. The coarse stage integrates TVCE, MPER and PEGR; OCRR performs the second-stage numerical correction. Both stages recover the incomplete input window.

3.2. Temporal–Variable Context Encoding (TVCE)

TVCE establishes a contextual representation for each target point by combining local temporal dynamics with cross-variable dependencies. We first standardize each variable using training-set statistics. A visible-only window normalization then supplies a mean μ_c and standard deviation σ_c, with training-statistic fallbacks when necessary. Masks are applied before numerical transformations. A residual temporal CNN processes variables with shared filters, using visible values, masks, temporal position and distances to visible observations. Global level features supplement this local representation.

Variable-signature attention exchanges information between entire temporal signatures. Following the channel–head geometry of T1 [6], each feature channel provides an attention pathway over variables. We use two temporal residual blocks and two variable-attention blocks. The resulting point features are pooled over overlapping patches:

$$\begin{aligned}\mathbf{F} &= \mathrm{VSAttn}\big(\mathrm{CNN}(\mathbf{X}_{\text{obs}}, \mathbf{M})\big),\\ z_{j,c} &= \phi_{\text{patch}}([P^x_{j,c}; P^m_{j,c}]) + \psi\big(\mathrm{Pool}(\mathbf{F}_{\mathcal{P}_j,c})\big).\end{aligned} \tag{2}$$

Here VSAttn denotes variable-signature attention, $\mathcal{P}_j$ is the jth patch, P^x, P^m are its visible values and mask, and ψ includes the projection of pooled context. We use $L = 96$, patch length 16, stride 8 and feature width 64. A base reconstruction head produces a vector $b_{j,c} \in \mathbb{R}^{16}$ from $z_{j,c}$.

3.3. Multi-Source Patch Evidence Retrieval (MPER)

MPER supplements the target context with identifiable fragments from complementary sources. For each damaged target patch, we construct $\mathcal{C}_{j,c}$ from three source families. Each member carries an encoded message, quality features, a

EviRefine architecture

TVCE + MPER
$\longrightarrow$ PEGR + confidence-weighted overlap fusion
$\longrightarrow$ OCRR

Figure 1. Overview of EviRefine. TVCE encodes point context, MPER constructs patch evidence, PEGR produces the coarse reconstruction, and OCRR refines it using visible observations.

source-type identity, a nonnegative within-source prior and an eligibility flag.

Temporal patch evidence. Candidates come from other patches of the same variable in the current window. Separate distance bands on either side organize near and distant context. The nearest band favors proximity; farther bands use a learned match. Patches without visible support are ineligible. The five bands in each direction yield at most ten candidates.

Cross-variable patch evidence. A window-dependent selector combines learned variable relationships with current source context to select at most four other variables. For each source, patches at offsets $-1, 0, 1$ provide up to twelve candidates. Visibility and agreement on commonly visible positions contribute to candidate quality. Candidate encoders include the target latent and contextual source representations.

Historical patch evidence. A training-only memory stores eligible patch members and retrieval keys. The query and key-supervision coordinates are aligned to the training-standardized scale; a locally normalized query is mapped back to this scale before key encoding. Retrieval respects source-variable and temporal eligibility, then returns up to four members from the selected prototype group. Member values are converted to the target's working scale for message construction. Published memory snapshots and their retrieval encoders are fixed during evaluation and refinement training.

The common bank retains individual patch members, their source identities and their support features. Its quality features and priors represent observable support and compatibility. Target conditioning is used in candidate construction as well as in the point query; the full pathway is evaluated against a context-only capacity control in Sec. 5.

3.4. Pointwise Evidence-Guided Reconstruction (PEGR)

PEGR converts patch evidence into point-specific corrections of the contextual base estimate. A target position forms a query q_p from its point context, patch latent $z_{j,c}$, visible features, relative position, base value and source availability. Candidate messages are normalized and augmented with projected quality features and source-type embeddings, then projected into keys and values. For eight heads of width $d_h = 8$, reading and residual prediction are

$$\begin{aligned} a_{p,k}^{(h)} &= \frac{\exp(\ell_{p,k}^{(h)})}{1 + \sum_{v \in \mathcal{C}_{j,c}} \exp(\ell_{p,v}^{(h)})}, \\ \ell_{p,k}^{(h)} &= \frac{\langle q_p^{(h)}, K_k^{(h)} \rangle}{\sqrt{d_h}} + \log \pi_k + \eta_k, \\ r_p &= \mathrm{Concat}_h \left(\sum_k a_{p,k}^{(h)} V_k^{(h)} \right), \\ \delta_p &= \mathbf{1}_{\exists k \in \mathcal{C}_{j,c}: \eta_k = 0}\, g_\theta([q_p; r_p]). \end{aligned} \tag{3}$$

where π_k is a source prior and η_k is zero for a legal candidate and $-\infty$ otherwise. A null entry with zero logit and zero value accounts for the constant 1 in the denominator. It attenuates the evidence readout when support is weak. The residual head jointly uses the query and concatenated readout; its correction is zero when no candidate is legal.

Patch estimates are converted to the training-standardized scale and merged by Confidence-Weighted Overlap Fusion using positive learned weights:

$$\begin{aligned} u_{j,p,c} &= \mu_c + \sigma_c b_{j,p,c} + s_c \delta_{j,p,c}, \\ s_c &= \sqrt{\sigma_c^2 + 0.1^2}, \\ w_{j,p,c} &= \exp\left(2 \tanh \frac{h_{j,p,c}}{2} \right), \\ X_{t,c}^{c} &= \frac{\sum_{j: t \in \mathcal{P}_j} w_{j,t-jS,c} u_{j,t-jS,c}}{\sum_{j: t \in \mathcal{P}_j} w_{j,t-jS,c}}. \end{aligned} \tag{4}$$

The numerical floor in s_c prevents a nearly constant visible window from collapsing the residual scale. $h_{j,p,c}$ is the output of the confidence network using point context, visible features, position, base/candidate values and source availability. These weights determine the contribution of each overlapping patch estimate. Padding is excluded from the sums and visible entries are restored after fusion.

3.5. Observation-Conditioned Residual Refinement (OCRR)

OCRR corrects the initial trajectory by combining temporal context with the numerical anchors supplied by visible observations. The second stage combines the fixed coarse estimate with original visible observations. At each variable and time, a reference $R_{t,c}$ is obtained by interpolation between nearest visible neighbors, or by the available neighbor when only one side exists. A support feature $U_{t,c}$ decreases with interpolation span or one-sided distance. Unavailable references have zero support.

The refiner projects five input groups into width 64, applies a local residual convolution and one temporal Transformer layer [8], and produces two corrections:

$$\begin{aligned} D &= U \odot (\mathbf{R} - \mathrm{sg}(\mathbf{X}^{\mathrm{c}})), \\ Z &= \mathrm{TemporalEncoder}\big([\mathrm{sg}(\mathbf{X}^{\mathrm{c}}); \mathbf{X}_{\mathrm{obs}}; \mathbf{M}; D; U]\big), \\ \Delta_{\mathrm{net}} &= A_{\mathrm{win}} \odot 2\tanh(W_n Z/2), \\ G &= A_{\mathrm{var}} \odot \tanh(W_g Z), \\ \widetilde{\mathbf{X}} &= \mathbf{X}^{\mathrm{c}} + \Delta_{\mathrm{net}} + G \odot (\mathbf{R} - \mathrm{sg}(\mathbf{X}^{\mathrm{c}})), \\ \widehat{\mathbf{X}} &= \mathbf{M} \odot \mathbf{X}_{\mathrm{obs}} + (\mathbf{1} - \mathbf{M}) \odot \widetilde{\mathbf{X}}. \end{aligned} \tag{5}$$

All numerical features in this equation use training-standardized coordinates. sg denotes stop-gradient; in implementation the entire coarse model is frozen. The activation indicators are $A_{\mathrm{win}} = \mathbf{1}[\sum_{t,c} M_{t,c} > 0]$ and $A_{\mathrm{var},c} = \mathbf{1}[\sum_t M_{t,c} > 0]$, broadcast over time and channels as appropriate. The first head predicts a bounded free residual, whereas the signed gate G controls the correction associated with the reference. Reference corrections are active only when that variable has visible support. Network corrections are disabled for wholly unobserved windows. The final mask operation exactly preserves visible values.

3.6. Training and Selection

We train coarse recovery first and then train the refiner for the fixed coarse checkpoint. The principal reconstruction terms on native-valid artificially hidden positions are

$$\begin{aligned} \mathcal{L}_{\mathrm{rec}}^{\mathrm{coarse}} &= \mathrm{MSE}_{\mathbf{E}}(\mathbf{X}^{\mathrm{c}}, \mathbf{X}), \\ \mathcal{L}_{\mathrm{rec}}^{\mathrm{refine}} &= \mathrm{MAE}_{\mathbf{E}}(\widehat{\mathbf{X}}, \mathbf{X}). \end{aligned} \tag{6}$$

Training combines these reconstruction terms with auxiliary regularization. Retrieval-key supervision aligns masked and complete keys and applies normalized contrastive supervision in training-global coordinates. Complete native-valid targets are used only as training labels. Both stages run 100 epochs with Adam, learning rate10^{-3}, zero weight decay, FP32 and batch 128. Coarse selection minimizes fixed-validation full-window MSE; refinement selection minimizes fixed-validation full-window MAE. The coarse network, memory members and retrieval encoders remain frozen throughout refinement training.

4. Experiments

4.1. Datasets and Evaluation Protocol

We evaluate EviRefine on ETTh1, Weather, Exchange-Rate, BeijingAirQuality, PhysioNet2012 and PAMAP2, covering industrial, meteorological, financial, environmental, clinical and wearable-sensor measurements. The prepared data use 7,21,8,12,35 and 40 variables, respectively. BeijingAirQuality treats the twelve stations as related channels. Native observation masks are retained; patient, subject and record boundaries are respected when constructing windows. All statistics used for scaling and history construction come from training data.

Four fixed missingness configurations combine block-start parameterp_s and block length b: $(.10, 8)$, $(.15, 8)$, $(.20, 8)$ and $(.10, 16)$. Realized missing coverage depends on the start parameter, overlapping blocks and boundary handling. All methods within a comparison receive identical held-out masks and are scored only at originally observed values that were artificially hidden.

We use three training seeds,2023,2024 and 2025, and twenty fixed test-mask realizations per trained model. For MSE, aggregation is

$$\begin{aligned} m_{s,k}^{\mathrm{MSE}} &= \frac{\sum_{t,c} E_{t,c}^{(k)} (\widehat{X}_{t,c}^{(s,k)} - X_{t,c})^2}{\sum_{t,c} E_{t,c}^{(k)}}, \\ \overline{m}_s &= \frac{1}{20} \sum_{k=1}^{20} m_{s,k}, \\ \overline{m} &= \frac{1}{3} \sum_{s \in \{2023,2024,2025\}} \overline{m}_s. \end{aligned} \tag{7}$$

MAE uses the corresponding absolute error. We report the mean and sample standard deviation of the three training-seed averages $\overline{m}_s$. Errors are measured in training-standardized coordinates. The main results and dedicated mechanism cohorts each use fixed, shared masks within their comparisons.

4.2. Formal Six-Dataset Results

Table 1 records the complete EviRefine evaluation across the six datasets and four masking configurations. It reports both MSE and MAE, enabling squared-error accuracy and absolute-error accuracy to be assessed together. The long-block and higher-start-rate cases test the model with reduced visible support. Dedicated paired studies below locate the contribution of patch evidence and refinement within this system.

4.3. Controlled Comparisons

The full candidate pathway is compared with a capacity-matched context-only model that retains the same front-end, base prediction, overlap fusion and refinement structure. Its

Dataset	p_s	b	MSE ↓	MAE ↓
ETTh1	.10	8	0.116781 ± 0.001319	0.205465 ± 0.001606
ETTh1	.15	8	0.172229 ± 0.001825	0.256546 ± 0.001479
ETTh1	.20	8	0.261890 ± 0.001713	0.322662 ± 0.001965
ETTh1	.10	16	0.334986 ± 0.006672	0.372039 ± 0.004057
Weather	.10	8	0.042431 ± 0.000577	0.043479 ± 0.000306
Weather	.15	8	0.052501 ± 0.000742	0.053946 ± 0.000418
Weather	.20	8	0.064593 ± 0.001175	0.070304 ± 0.000500
Weather	.10	16	0.078572 ± 0.000796	0.092897 ± 0.000638
Exchange	.10	8	0.009212 ± 0.002400	0.057764 ± 0.012104
Exchange	.15	8	0.015368 ± 0.002077	0.073880 ± 0.007749
Exchange	.20	8	0.054194 ± 0.001231	0.098458 ± 0.001036
Exchange	.10	16	0.162275 ± 0.004526	0.172044 ± 0.001131
Beijing AQ	.10	8	0.077236 ± 0.000584	0.144566 ± 0.000524
Beijing AQ	.15	8	0.094839 ± 0.000677	0.163089 ± 0.000209
Beijing AQ	.20	8	0.122833 ± 0.001111	0.188204 ± 0.000303
Beijing AQ	.10	16	0.146832 ± 0.000833	0.206913 ± 0.000600
PhysioNet	.10	8	0.386760 ± 0.003590	0.343365 ± 0.001006
PhysioNet	.15	8	0.476532 ± 0.001034	0.403849 ± 0.000135
PhysioNet	.20	8	0.564794 ± 0.000846	0.455283 ± 0.000148
PhysioNet	.10	16	0.581826 ± 0.001249	0.469871 ± 0.000192
PAMAP2	.10	8	0.227562 ± 0.002636	0.181577 ± 0.001163
PAMAP2	.15	8	0.326661 ± 0.001309	0.230819 ± 0.000952
PAMAP2	.20	8	0.466878 ± 0.004067	0.292715 ± 0.000651
PAMAP2	.10	16	0.532003 ± 0.003002	0.334631 ± 0.001549

Table 1. Formal EviRefine results over six datasets. Each cell is the mean±sample SD over three training seeds, after averaging twenty shared test masks per model. p_s is the block-start parameter and b is the block length. The table reports measured EviRefine performance across all configurations.

contextual recovery network uses an analytically chosen hidden width to match effective inference parameters within 1%; all added parameters participate in prediction. The context-only model is independently trained with the same data, masks, optimizer budget and checkpoint-selection criteria.

We further remove temporal, cross-variable or historical evidence individually; replace selection with source-wise random legal candidates; use a unified visible-value similarity search; or share one read across each target patch. Random and unified selection match the reference effective candidate budget. Other structures and the refinement training protocol remain fixed. Source removals measure conditional gains of one pathway within the complete system.

An additional BeijingAirQuality study applies the same refinement architecture and training budget to T1, the matched context-only model and EviRefine. It separates raw backbone outputs from outputs receiving a common second-stage correction. Backbone weights stay frozen during this comparison.

4.4. Implementation

The main width is 64, with eight point-readout heads and one width 64, four-head temporal refinement layer. Patch length and stride are 16 and 8. Each imputation stage runs 100 epochs; there is no training early stop in these EviRefine comparisons. The first stage is selected by validation MSE, and the refiner by validation MAE. No test metric is used for checkpoint selection.

For efficiency, we time fixed batch 128 inputs on a single exclusive RTX5090 in FP32, using twenty warm-up iterations and one hundred synchronized measurements. We report median latency, effective parameter count and peak allocated GPU memory. Training-loop time includes validation and saving; it is recorded separately from full experiment wall time.

5. Model Analysis

5.1. Value of the Complete Evidence Pathway

Table 2 compares EviRefine and the matched context-only model in dedicated paired cohorts: ETTh1$(.10, 16)$ and PAMAP2$(.15, 8)$. All three training-seed pairs favor EviRefine in both final MSE and MAE. The final MSE reductions are 5.19% and 2.39%, while MAE reductions are 3.52% and 1.66%, respectively. The coarse outputs already exhibit gains, and the differences remain after independently training the same refiner. This establishes an incremental value

of explicit patch evidence beyond the matched contextual alternative.

5.2. Source Organization and Pointwise Reading

Table 3 shows that removing any one source increases mean final MSE in the two analyzed datasets. Cross-variable evidence has the largest source-removal MSE effect on ETTh1, whereas temporal evidence has the largest effect on PAMAP2. Their different contributions are consistent with source usefulness depending on the underlying signals.

Source-wise random selection and unified retrieval produce higher mean MSE than the complete system. Retaining provenance-specific organization therefore provides benefits beyond merely supplying the same number of legal fragments. Patch-shared reading changes the accuracy tradeoff: it gives lower ETTh1 MSE, while pointwise reading gives lower ETTh1 MAE and PAMAP2 MSE. We assess reading granularity using both error criteria.

5.3. Observation-Conditioned Refinement

For the paired EviRefine cohorts, refinement reduces MSE from 0.384873 to 0.334986 on ETTh1 and from 0.336419 to 0.324572 on PAMAP2, corresponding to 12.96% and 3.52%. The network combines temporal correction with reference information anchored to visible observations. Coordinate-level diagnostics separately account for error reductions and increases, making the aggregate correction traceable to its contributing positions.

The common-refiner study in Table 4 further shows its effect across backbones. On BeijingAirQuality$(.20, 8)$, T1 changes from 0.225523 to 0.129252 MSE after refinement; EviRefine changes from 0.179570 to 0.123941. The common correction stage accounts for a substantial part of the raw-output gap and preserves an additional advantage for EviRefine's coarse estimate. This comparison attributes the backbone and refinement effects separately.

5.4. Missing-Condition Diagnostics

We group evaluation coordinates by mask-defined gap length, nearest visible-point distance, patch visibility and window boundary. These groups are fixed before reading model outputs. On ETTh1$(.10, 16)$, gains remain positive across all three training seeds for coordinates 17–32 and 33–95 steps from visible observations, supporting the use of patch evidence where immediate observations are sparse. For coordinates whose overlapping patches contain no visible observations, the external-to-patch context and other evidence sources still support recovery. The full stratified values and coordinate support are maintained with the numerical evidence tables.

5.5. Accuracy and Cost

Table 5 quantifies the additional cost of the candidate pathway. EviRefine and the matched context-only model have nearly equal effective parameter counts, while candidate construction and reading increase latency and allocated memory. On PAMAP2, the matched models use 575074 and 574794 effective parameters, with full-pipeline p50 latency 98.48 and 55.53ms per batch. This makes the measured accuracy gain assessable alongside the execution cost.

Readout sensitivity. With total width 64 fixed, Table 6 varies the number of point-readout heads. Parameter count is unchanged, isolating how the feature width is partitioned. The primary model retains its predefined eight-head configuration; the sensitivity study uses seed 2023 and validation-selected checkpoints.

Dataset	Model	Coarse MSE	Refined MSE	Refined MAE
ETTh1	EviRefine	0.384873 ± 0.006236	0.334986 ± 0.006672	0.372039 ± 0.004057
ETTh1	Context-only	0.408733 ± 0.001863	0.353320 ± 0.009562	0.385618 ± 0.010399
PAMAP2	EviRefine	0.336419 ± 0.000680	0.324572 ± 0.001591	0.230532 ± 0.000622
PAMAP2	Context-only	0.351320 ± 0.002814	0.332533 ± 0.001960	0.234430 ± 0.001031

Table 2. Paired evidence-pathway analysis. Both datasets use three training seeds and twenty shared test masks per model. The PAMAP2 mechanism cohort is separate from its formal benchmark cohort in Table 1. Coarse and refined outputs come from the same selected model pair.

Variant	ETTh1 MSE	ETTh1 MAE	PAMAP2 MSE	PAMAP2 MAE
EviRefine	0.334986 ± 0.006672	0.372039 ± 0.004057	0.322915	0.230183
w/o temporal evidence	0.343612 ± 0.006384	0.381684 ± 0.004346	0.333096	0.235096
w/o variable evidence	0.349493 ± 0.018407	0.381209 ± 0.010604	0.325663	0.231132
w/o historical evidence	0.339364 ± 0.004246	0.376169 ± 0.002895	0.324277	0.231290
Source-random	0.336177 ± 0.006092	0.377530 ± 0.004732	0.325863	0.231077
Unified retrieval	0.339484 ± 0.005147	0.380305 ± 0.002030	0.331974	0.232928
Context-only (matched)	0.353320 ± 0.009562	0.385618 ± 0.010399	0.334383	0.235483
Patch-shared read	0.333639 ± 0.004381	0.374103 ± 0.002174	0.323958	0.229965

Table 3. Component and retrieval controls, after refinement. ETTh1 uses three training seeds (mean±sample SD); PAMAP2 component rows consistently use seed 2023. Every model is evaluated on twenty shared masks. Three-seed PAMAP2 results for EviRefine and the matched context-only model are given separately in Table 2.

Backbone	Coarse MSE	Refined MSE	Coarse MAE	Refined MAE
T1	0.225523	0.129252	0.264340	0.193203
Context-only	0.188083	0.124760	0.243393	0.189678
EviRefine	0.179570	0.123941	0.235067	0.188432

Table 4. A common refinement architecture applied to three frozen backbones on BeijingAirQuality$(.20, 8)$, seed 2023, twenty test masks. Coarse and refined errors are evaluated on the same positions.

Dataset	Model	Params	p50 (ms)	VRAM (GiB)
PAMAP2	EviRefine	575,074	98.48	6.121
PAMAP2	Context-only	574,794	55.53	4.122
ETTh1	EviRefine	353,248	17.63	1.067
ETTh1	Context-only	352,968	11.51	0.709

Table 5. Inference cost of selected seed 2023 models on the same exclusive RTX5090, FP32, batch 128. Median latency follows twenty warm-ups and one hundred synchronized measurements. VRAM denotes peak allocated memory.

Heads	Val MSE	Val MAE	Test MSE	Test MAE
4	0.422429	0.427810	0.326837	0.370128
8	0.446899	0.436511	0.336727	0.373539
16	0.425259	0.428136	0.325889	0.369931

Table 6. Point-readout head sensitivity on ETTh1$(.10, 16)$, width 64, seed 2023 and twenty shared test masks.

6. Conclusion

We introduced EviRefine, a coarse-to-fine multivariate time-series imputer that combines a strong contextual estimate with explicit multi-source patch evidence and observation-conditioned residual refinement. Paired capacity controls identify an accuracy contribution from the full evidence pathway that remains after refinement. A common-refiner study separates backbone quality from second-stage correction. Together with source, reading-granularity and cost analyses, these results characterize how evidence organization and numerical refinement contribute to recovery.

LLM Usage Statement

OpenAI Codex was used to assist implementation, experiment orchestration, numerical-table preparation, literature organization, and English/Chinese drafting and editing. The authors are responsible for the method, experimental protocol, numerical sources, references and final scientific claims.

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